



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

SERVICE DISCOVERY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Microservices

TOTAL: 10 QUESTIONS

Q1 What is Service Discovery and what problem in Microservices does it solve?

Q6 How does Service Discovery handle reliability and high availability?

Q2 What are the core components or concepts in Service Discovery?

Q7 What observability does Service Discovery provide out of the box?

Q3 How does Service Discovery compare to its main alternatives?

Q8 What are common performance bottlenecks in Service Discovery?

Q4 How do you get started with Service Discovery for a new project?

Q9 What security best practices apply when running Service Discovery?

Q5 What configuration options most affect Service Discovery's behavior in production?

Q10 What mistakes do teams commonly make when adopting Service Discovery?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Service Discovery is a tool or platform

Mitigation

Service Discovery is a tool or platform that automates or simplifies a specific concern in the Microservices domain, reducing manual effort and operational complexity for engineering teams.

A6 Service Discovery provides HA through leader election, replication, or stateless

Availability

Service Discovery provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Service Discovery has key building blocks that

Primitives

Service Discovery has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Service Discovery exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

Observation

A3 Service Discovery trades off simplicity against feature richness differently than its alternatives. Choose Service Discovery when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

Hardening

Service Discovery trades off simplicity against feature richness differently than its alternatives. Choose Service Discovery when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

Performance

A4 Install Service Discovery via its package manager

Gotchas

Install Service Discovery via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Service Discovery with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

Assessment

Run Service Discovery with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.